Status Report #4 - Romano

2025-10-16

Setup

```r
# Load in libraries
library(MASS)
library(e1071)
library(rpart)
library(randomForest)
```

```
## randomForest 4.7-1.2

## Type rfNews() to see new features/changes/bug fixes.
```

```r
# Load data and view it (make sure everything looks right)
HeartAttack <- read.csv("HeartAttack.csv")
head(HeartAttack)
```

```
##   Age Gender Cholesterol BloodPressure HeartRate  BMI Smoker Diabetes
## 1  31   Male         194           162        71 22.9      0        1
## 2  69   Male         208           148        93 33.9      1        1
## 3  34 Female         132           161        94 34.0      0        0
## 4  53   Male         268           134        91 35.0      0        1
## 5  57 Female         203           140        75 30.1      0        1
## 6  41   Male         158           154        72 38.7      0        1
##   Hypertension FamilyHistory PhysicalActivity AlcoholConsumption      Diet
## 1            0             0                6                  0 Unhealthy
## 2            0             0                1                  2 Unhealthy
## 3            1             1                1                  3   Healthy
## 4            1             0                6                  0   Healthy
## 5            0             0                4                  1  Moderate
## 6            0             1                3                  2  Moderate
##   StressLevel Ethnicity Income EducationLevel Medication ChestPainType
## 1           1  Hispanic  64510    High School        Yes       Typical
## 2           6     Asian  91773        College         No      Atypical
## 3           3     Black 173550        College         No   Non-anginal
## 4           3  Hispanic  43861    High School        Yes      Atypical
## 5           1  Hispanic  83404    High School        Yes       Typical
## 6           4     Other 113011    High School        Yes   Non-anginal
##         ECGResults MaxHeartRate ST_Depression ExerciseInducedAngina       Slope
## 1 ST-T abnormality          173          0.52                   Yes Downsloping
## 2   LV hypertrophy          189          3.79                   Yes   Upsloping
## 3           Normal          122          0.17                   Yes   Upsloping
## 4 ST-T abnormality          104          0.67                   Yes        Flat
## 5 ST-T abnormality          126          5.00                   Yes        Flat
## 6   LV hypertrophy          155          4.30                    No Downsloping
##   NumberOfMajorVessels       Thalassemia PreviousHeartAttack StrokeHistory
## 1                    1            Normal                   0             0
## 2                    2            Normal                   0             0
```

```
## 3                    0            Normal                  1             0
## 4                    0 Reversible defect                  1             0
## 5                    0      Fixed defect                  1             0
## 6                    2 Reversible defect                  0             0
##   Residence EmploymentStatus MaritalStatus         Outcome
## 1  Suburban          Retired        Single No Heart Attack
## 2  Suburban       Unemployed       Married No Heart Attack
## 3     Rural          Retired        Single    Heart Attack
## 4  Suburban          Retired       Widowed No Heart Attack
## 5     Rural          Retired       Married    Heart Attack
## 6     Rural       Unemployed       Married    Heart Attack
```

```r
# Load cleaned data from Status Report #1 and view it (make sure it looks right)
HeartAttackClean <- readRDS("HeartAttackClean.rds")
head(HeartAttackClean)
```

```
##   Age Gender Cholesterol BloodPressure HeartRate  BMI Smoker Diabetes
## 1  31      0         194           162        71 22.9      0        1
## 2  69      0         208           148        93 33.9      1        1
## 3  34      1         132           161        94 34.0      0        0
## 4  53      0         268           134        91 35.0      0        1
## 5  57      1         203           140        75 30.1      0        1
## 6  41      0         158           154        72 38.7      0        1
##   Hypertension FamilyHistory PhysicalActivity AlcoholConsumption Diet
## 1            0             0                6                  0    0
## 2            0             0                1                  2    0
## 3            1             1                1                  3    1
## 4            1             0                6                  0    1
## 5            0             0                4                  1    2
## 6            0             1                3                  2    2
##   StressLevel Ethnicity Income EducationLevel Medication ChestPainType
## 1           1         0  64510              0          0             0
## 2           6         1  91773              1          1             1
## 3           3         2 173550              1          1             2
## 4           3         0  43861              0          0             1
## 5           1         0  83404              0          0             0
## 6           4         3 113011              0          0             2
##   ECGResults MaxHeartRate ST_Depression ExerciseInducedAngina Slope
## 1          0          173          0.52                     0     0
## 2          1          189          3.79                     0     1
## 3          2          122          0.17                     0     1
## 4          0          104          0.67                     0     2
## 5          0          126          5.00                     0     2
## 6          1          155          4.30                     1     0
##   NumberOfMajorVessels Thalassemia PreviousHeartAttack StrokeHistory Residence
## 1                    1           0                   0             0         0
## 2                    2           0                   0             0         0
## 3                    0           0                   1             0         1
## 4                    0           1                   1             0         0
## 5                    0           2                   1             0         1
## 6                    2           1                   0             0         1
##   EmploymentStatus MaritalStatus Outcome
## 1                0             0       0
## 2                1             1       0
## 3                0             0       1
```

```
## 4                0              2       0
## 5                0              1       1
## 6                1              1       1
```

Logistic Regression Model - Decision Tree

```r
# Choose predictors
predictors <- c("Age","Gender","FamilyHistory","Ethnicity",
                "Smoker","BloodPressure","Cholesterol","BMI",
                "Diabetes","Diet","PhysicalActivity","HeartRate",
                "AlcoholConsumption")

# Split into training and testing sets 70/30
set.seed(1)
train_idx <- sample(1:nrow(HeartAttackClean), 0.7 * nrow(HeartAttackClean))
train <- HeartAttackClean[train_idx, ]
test  <- HeartAttackClean[-train_idx, ]

# Fit logistic regression
fit <- glm(
  reformulate(intersect(predictors, names(HeartAttackClean)), response = "Outcome"),
  data = HeartAttackClean,
  family = binomial(),
  na.action = na.omit
)

# Add decision tree enhancement
tree_fit <- rpart(Outcome ~ ., data = train, method = "class"); tree_pred <- predict(tree_fit, newdata =

# Results
summary(fit)          # coefficients (log-odds), p-values
```

```
##
## Call:
## glm(formula = reformulate(intersect(predictors, names(HeartAttackClean)),
##     response = "Outcome"), family = binomial(), data = HeartAttackClean,
##     na.action = na.omit)
##
## Coefficients:
##                      Estimate Std. Error z value Pr(>|z|)
## (Intercept)         1.750e-02  3.461e-02   0.506   0.6131
## Age                -1.201e-04  2.064e-04  -0.582   0.5608
## Gender              7.578e-03  6.550e-03   1.157   0.2473
## FamilyHistory       1.638e-03  6.550e-03   0.250   0.8025
## Ethnicity          -3.912e-03  2.315e-03  -1.690   0.0911 .
## Smoker              3.328e-03  6.550e-03   0.508   0.6114
## BloodPressure       3.769e-05  1.260e-04   0.299   0.7649
## Cholesterol        -6.805e-05  5.674e-05  -1.199   0.2304
## BMI                 1.336e-04  5.161e-04   0.259   0.7957
## Diabetes            2.905e-03  6.550e-03   0.443   0.6574
## Diet                1.606e-03  4.009e-03   0.401   0.6887
## PhysicalActivity    3.365e-04  1.637e-03   0.205   0.8372
## HeartRate          -1.693e-04  1.888e-04  -0.896   0.3700
## AlcoholConsumption  2.387e-03  2.315e-03   1.031   0.3025
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 517051  on 372973  degrees of freedom
## Residual deviance: 517043  on 372960  degrees of freedom
## AIC: 517071
##
## Number of Fisher Scoring iterations: 3
```

```
exp(coef(fit))        # odds ratios
```

```
##        (Intercept)                Age             Gender      FamilyHistory
##          1.0176503          0.9998799          1.0076068          1.0016396
##          Ethnicity             Smoker      BloodPressure        Cholesterol
##          0.9960957          1.0033337          1.0000377          0.9999320
##                BMI           Diabetes               Diet   PhysicalActivity
##          1.0001336          1.0029090          1.0016073          1.0003365
##          HeartRate AlcoholConsumption
##          0.9998307          1.0023899
```

```
# Simple in-sample predictions & accuracy (threshold = 0.5)
p    <- predict(fit, type = "response")
pred <- as.integer(p >= 0.5)
table(truth = HeartAttackClean$Outcome, pred = pred)
```

```
##      pred
## truth      0      1
##     0 107199  79459
##     1 106080  80236
```

```
mean(pred == HeartAttackClean$Outcome)  # accuracy
```

```
## [1] 0.5025417
```

```
mean(tree_pred == test$Outcome)
```

```
## [1] 0.500076
```

Logistic Regression Model - Random Forest

```
# Choose predictors
predictors <- c("Age","Gender","FamilyHistory","Ethnicity",
                "Smoker","BloodPressure","Cholesterol","BMI",
                "Diabetes","Diet","PhysicalActivity","HeartRate",
                "AlcoholConsumption")

# Split into training and testing sets (70/30 for example)
set.seed(1)
train_idx <- sample(1:nrow(HeartAttackClean), 0.7 * nrow(HeartAttackClean))
train <- HeartAttackClean[train_idx, ]
test  <- HeartAttackClean[-train_idx, ]

# Fit logistic regression
fit <- glm(
  reformulate(intersect(predictors, names(HeartAttackClean)), response = "Outcome"),
  data = HeartAttackClean,
  family = binomial(),
```

```r
  na.action = na.omit
)

# Results
summary(fit)          # coefficients (log-odds), p-values
```

```
##
## Call:
## glm(formula = reformulate(intersect(predictors, names(HeartAttackClean)),
##     response = "Outcome"), family = binomial(), data = HeartAttackClean,
##     na.action = na.omit)
##
## Coefficients:
##                        Estimate Std. Error z value Pr(>|z|)
## (Intercept)           1.750e-02  3.461e-02   0.506   0.6131
## Age                  -1.201e-04  2.064e-04  -0.582   0.5608
## Gender                7.578e-03  6.550e-03   1.157   0.2473
## FamilyHistory         1.638e-03  6.550e-03   0.250   0.8025
## Ethnicity            -3.912e-03  2.315e-03  -1.690   0.0911 .
## Smoker                3.328e-03  6.550e-03   0.508   0.6114
## BloodPressure         3.769e-05  1.260e-04   0.299   0.7649
## Cholesterol          -6.805e-05  5.674e-05  -1.199   0.2304
## BMI                   1.336e-04  5.161e-04   0.259   0.7957
## Diabetes              2.905e-03  6.550e-03   0.443   0.6574
## Diet                  1.606e-03  4.009e-03   0.401   0.6887
## PhysicalActivity      3.365e-04  1.637e-03   0.205   0.8372
## HeartRate            -1.693e-04  1.888e-04  -0.896   0.3700
## AlcoholConsumption    2.387e-03  2.315e-03   1.031   0.3025
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 517051  on 372973  degrees of freedom
## Residual deviance: 517043  on 372960  degrees of freedom
## AIC: 517071
##
## Number of Fisher Scoring iterations: 3
```

```r
exp(coef(fit))        # odds ratios
```

```
##        (Intercept)                Age             Gender      FamilyHistory
##          1.0176503          0.9998799          1.0076068          1.0016396
##          Ethnicity             Smoker      BloodPressure        Cholesterol
##          0.9960957          1.0033337          1.0000377          0.9999320
##                BMI           Diabetes               Diet   PhysicalActivity
##          1.0001336          1.0029090          1.0016073          1.0003365
##          HeartRate AlcoholConsumption
##          0.9998307          1.0023899
```

```r
# Simple in-sample predictions & accuracy (threshold = 0.5)
p    <- predict(fit, type = "response")
pred <- as.integer(p >= 0.5)
table(truth = HeartAttackClean$Outcome, pred = pred)
```

```
##      pred
## truth      0      1
##     0 107199  79459
##     1 106080  80236
```

```r
mean(pred == HeartAttackClean$Outcome)  # accuracy
```

```
## [1] 0.5025417
```

```r
# Add random forest enhancement
#rf_fit <- randomForest(Outcome ~ ., data = na.omit(train), ntree = 500)
#rf_pred <- predict(rf_fit, newdata = na.omit(test))
## I need some help with this. It keeps buffering, and it will not load results
```

LDA Model - Decsion Tree

```r
# Choose predictors
predictors <- c("Age","Gender","FamilyHistory","Ethnicity",
                "Smoker","BloodPressure","Cholesterol","BMI",
                "Diabetes","Diet","PhysicalActivity","HeartRate",
                "AlcoholConsumption")


# Split into training and test sets
set.seed(1)
n <- nrow(HeartAttackClean)
id <- sample(n, floor(0.8 * n))
train <- HeartAttackClean[id, ]
test  <- HeartAttackClean[-id, ]

# Fit LDA model
lda_fit <- lda(Outcome ~ Age + Gender + FamilyHistory + Ethnicity +
               Smoker + BloodPressure + Cholesterol + BMI + Diabetes +
               Diet + PhysicalActivity + HeartRate + AlcoholConsumption,
               data = train)

# Add decision tree enhancement
tree_fit <- rpart(Outcome ~ ., data = train, method = "class"); tree_pred <- predict(tree_fit, newdata =


# Predict on test data
lda_pred <- predict(lda_fit, newdata = test)$class

# Confusion matrix and accuracy
lda_cm <- table(truth = test$Outcome, pred = lda_pred)
lda_acc <- mean(lda_pred == test$Outcome)

lda_cm
```

```
##      pred
## truth     0     1
##     0 22307 14967
##     1 22374 14947
```

```r
lda_acc
```

```
## [1] 0.4994169
```

```r
mean(tree_pred == test$Outcome)
```

```
## [1] 0.499685
```

LDA Model - Random Forest

```r
# Choose predictors
predictors <- c("Age","Gender","FamilyHistory","Ethnicity",
                "Smoker","BloodPressure","Cholesterol","BMI",
                "Diabetes","Diet","PhysicalActivity","HeartRate",
                "AlcoholConsumption")


# Split into training and test sets
set.seed(1)
n <- nrow(HeartAttackClean)
id <- sample(n, floor(0.8 * n))
train <- HeartAttackClean[id, ]
test  <- HeartAttackClean[-id, ]

# Fit LDA model
lda_fit <- lda(Outcome ~ Age + Gender + FamilyHistory + Ethnicity +
               Smoker + BloodPressure + Cholesterol + BMI + Diabetes +
               Diet + PhysicalActivity + HeartRate + AlcoholConsumption,
               data = train)

# Add random forest enhancement
#rf_fit <- randomForest(Outcome ~ ., data = train, ntree = 500, importance = TRUE); rf_pred <- predict(
## Same issue as above

# Predict on test data
lda_pred <- predict(lda_fit, newdata = test)$class

# Confusion matrix and accuracy
lda_cm <- table(truth = test$Outcome, pred = lda_pred)
lda_acc <- mean(lda_pred == test$Outcome)

lda_cm
```

```
##      pred
## truth     0     1
##     0 22303 14971
##     1 22370 14951
```

```r
lda_acc
```

```
## [1] 0.4994169
```

QDA Model - Decision Tree

```r
# Choose predictors
predictors <- c("Age","Gender","FamilyHistory","Ethnicity",
                "Smoker","BloodPressure","Cholesterol","BMI",
                "Diabetes","Diet","PhysicalActivity","HeartRate",
                "AlcoholConsumption")

set.seed(1)
```

```r
n  <- nrow(HeartAttackClean)
id <- sample(n, floor(0.8*n))
train <- HeartAttackClean[id, ]
test  <- HeartAttackClean[-id, ]

train  <- train[complete.cases(train[, c("Outcome", predictors)]), ]
test_cc <- test[complete.cases(test[, predictors]), ]

qda_fit  <- qda(as.formula(paste("Outcome ~", paste(predictors, collapse = " + "))), data = train)
qda_pred <- predict(qda_fit, newdata = test_cc)$class

# Add decsion tree enhancement
tree_fit <- rpart(Outcome ~ ., data = train, method = "class"); tree_pred <- predict(tree_fit, newdata =

qda_cm  <- table(truth = test_cc$Outcome, pred = qda_pred)
qda_acc <- mean(qda_pred == test_cc$Outcome)

qda_cm
```

```
##      pred
## truth     0     1
##     0 19730 17544
##     1 19698 17623
```

```r
qda_acc
```

```
## [1] 0.500744
```

```r
mean(tree_pred == test$Outcome)
```

```
## [1] 0.499685
```

QDA Model - Random Forest

```r
# Choose predictors
predictors <- c("Age","Gender","FamilyHistory","Ethnicity",
                "Smoker","BloodPressure","Cholesterol","BMI",
                "Diabetes","Diet","PhysicalActivity","HeartRate",
                "AlcoholConsumption")

set.seed(1)
n  <- nrow(HeartAttackClean)
id <- sample(n, floor(0.8*n))
train <- HeartAttackClean[id, ]
test  <- HeartAttackClean[-id, ]

train  <- train[complete.cases(train[, c("Outcome", predictors)]), ]
test_cc <- test[complete.cases(test[, predictors]), ]

qda_fit  <- qda(as.formula(paste("Outcome ~", paste(predictors, collapse = " + "))), data = train)
qda_pred <- predict(qda_fit, newdata = test_cc)$class

# Add random forest enhancement

qda_cm  <- table(truth = test_cc$Outcome, pred = qda_pred)
qda_acc <- mean(qda_pred == test_cc$Outcome)
```

```r
qda_cm
```

```
##      pred
## truth     0     1
##     0 19730 17544
##     1 19698 17623
```

```r
qda_acc
```

```
## [1] 0.500744
```

Naive Bayes Model - Decision Tree

```r
# Choose predictors
predictors <- c("Age","Gender","FamilyHistory","Ethnicity",
                "Smoker","BloodPressure","Cholesterol","BMI",
                "Diabetes","Diet","PhysicalActivity","HeartRate",
                "AlcoholConsumption")
set.seed(1)
n  <- nrow(HeartAttackClean)
id <- sample(n, floor(0.8*n))
train <- HeartAttackClean[id, ]
test  <- HeartAttackClean[-id, ]

train  <- train[complete.cases(train[, c("Outcome", predictors)]), ]
test_cc <- test[complete.cases(test[, predictors]), ]

nb_fit  <- naiveBayes(as.formula(paste("Outcome ~", paste(predictors, collapse = " + "))), data = train
nb_pred <- predict(nb_fit, newdata = test_cc, type = "class")

# Add decision tree enhancement
tree_fit <- rpart(Outcome ~ ., data = train, method = "class"); tree_pred <- predict(tree_fit, newdata =

nb_cm  <- table(truth = test_cc$Outcome, pred = nb_pred)
nb_acc <- mean(nb_pred == test_cc$Outcome)

nb_cm
```

```
##      pred
## truth     0     1
##     0 22307 14967
##     1 22290 15031
```

```r
nb_acc
```

```
## [1] 0.5005429
```

```r
mean(tree_pred == test$Outcome)
```

```
## [1] 0.499685
```

Naive Bayes Model - Random Forest

```r
test$Outcome <- as.factor(test$Outcome)
# Choose predictors
predictors <- c("Age","Gender","FamilyHistory","Ethnicity",
                "Smoker","BloodPressure","Cholesterol","BMI",
                "Diabetes","Diet","PhysicalActivity","HeartRate",
```

```r
                 "AlcoholConsumption")
set.seed(1)
n  <- nrow(HeartAttackClean)
id <- sample(n, floor(0.8*n))
train <- HeartAttackClean[id, ]
test  <- HeartAttackClean[-id, ]

train  <- train[complete.cases(train[, c("Outcome", predictors)]), ]
test_cc <- test[complete.cases(test[, predictors]), ]

nb_fit  <- naiveBayes(as.formula(paste("Outcome ~", paste(predictors, collapse = " + "))), data = train)
nb_pred <- predict(nb_fit, newdata = test_cc, type = "class")

# Add random forest enhancement

nb_cm  <- table(truth = test_cc$Outcome, pred = nb_pred)
nb_acc <- mean(nb_pred == test_cc$Outcome)

nb_cm
```

```
##      pred
## truth     0     1
##     0 22307 14967
##     1 22290 15031
```

```r
nb_acc
```

```
## [1] 0.5005429
```

Conclusion:
- Adding decsion trees to my models did not improve the accuracy as I was hoping it would
- I was looking for a durastic difference in accuracy percentage, but there was little to noo change
- I am hopeful that adding a random forest enhancement will improve my accuracy, but I was not able to get that to work with my data after a lot of research